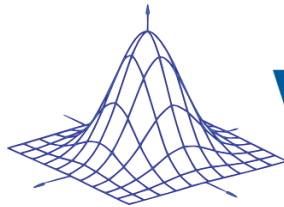


Home Energy Optimisation: Enhancing Heat Pump Efficiency through Data-Driven Insights

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WCPM

CATAPULT
Energy Systems

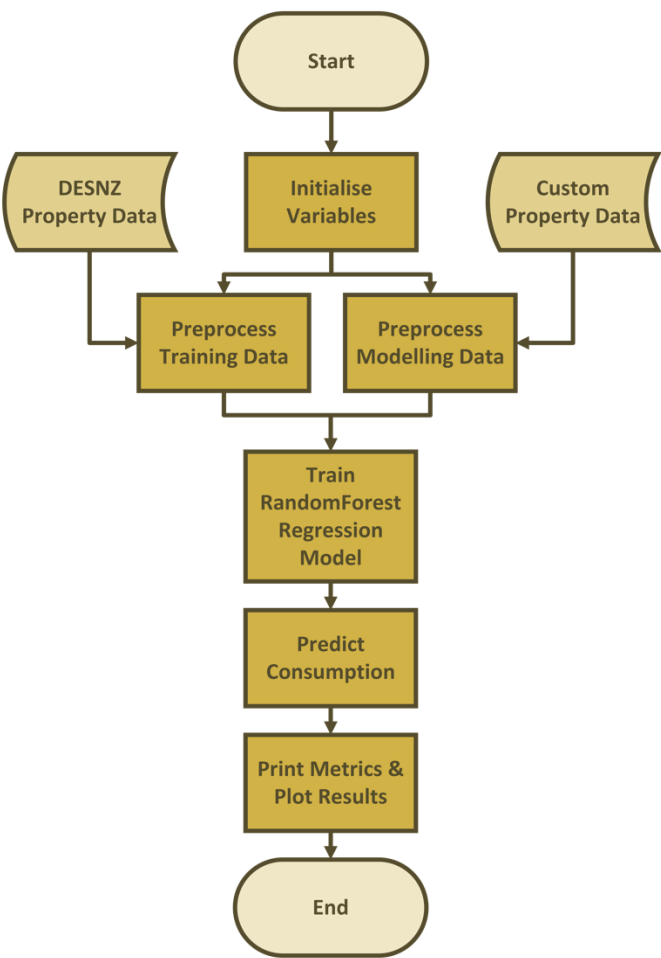
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Motivation

The Department for Energy Security and Net Zero (DESNZ) has launched the "Electrification of Heat" project to decarbonise domestic heating through air source heat pumps (ASHPs). This study uses DESNZ and Energy Systems Catapult datasets and Python scientific computing methodologies to develop machine learning models that predict ASHP energy consumption, efficiency, and cost savings. The goal is to support informed decisions by policymakers, consumers, and energy providers, ultimately reducing electricity costs and carbon emissions in UK households.

Methodology



The program uses data from **two sources**:

- DESNZ Property Dataset^[1]
- Custom Property

Preprocessing the datasets involves **cleaning, transforming, and normalising** for input into the machine learning model.

A RandomForestRegressor^[2] model is **trained** using the training data and utilised to **predict** energy consumption for the Custom Property.

Finally, performance metrics are printed to evaluate the accuracy and efficacy of the model, as well as **plotted results** for visual analysis.

This approach ensures that the model is both trained effectively and evaluated rigorously using standard metrics.

Preprocessing Calculations

Coefficient of Performance (COP)

a metric measuring the energy efficiency for an ASHP, and in this case, heating at a specific temperature^[3]

$$COP = \frac{\text{Heat Pump Energy Output}_{ij}}{\text{Whole System Energy Consumed}_{ij}}$$

Seasonal Coefficient of Performance (SCOP)

a metric measuring energy efficiency while taking seasonal temperature variations into account using partial load ratios^[4]

$$SCOP = \frac{\sum_i \left(\frac{1}{N_i} \sum_{j=1}^{N_i} \frac{\text{Heat Pump Energy Output}_{ij}}{\text{Heat Pump Energy Output}_{ij} + \text{Auxiliary Power}_{ij}} \times \text{Partial Load Ratio}_i \right)}{\sum_i \text{Partial Load Ratio}_i}$$

Internal Air Temperature [°C]	External Air Temperature [°C]	Partial Load Ratio [%]
20	-7	88
20	2	54
20	7	35
20	12	15

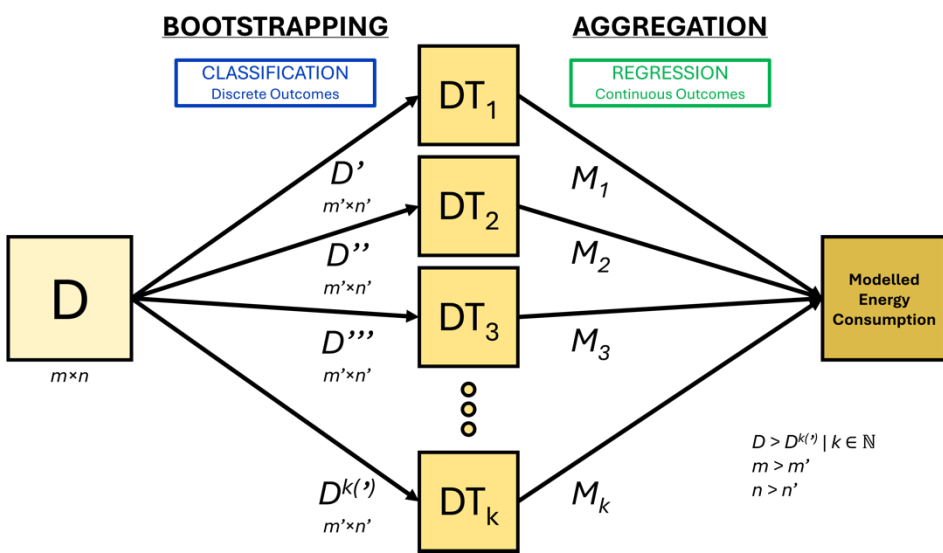
Modelling

RandomForestRegressor

a machine learning model for regression tasks, used to predict continuous numerical values by building an ensemble of decision trees and averaging their predictions for a final output^[2]

Components

- **Decision Tree:** Each node is a feature test, branches are test outcomes, and leaves are predicted values.
- **Ensemble Learning:** Trees work together to form a random forest, improving prediction accuracy.
- **Random Forest:** Each tree is trained on a random subset of features, improving robustness.



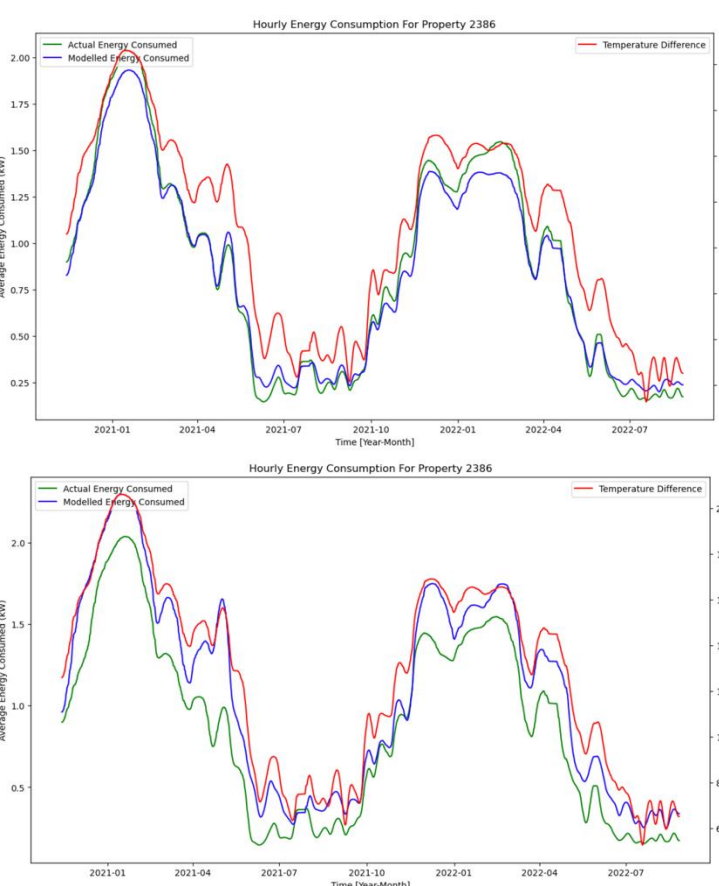
How It Works

1. **Bagging (Bootstrap Aggregating):** Multiple trees trained on random samples.
2. **Feature Selection:** At each split, a random features subset is considered to reduce correlation between trees.
3. **Tree Building:** Growth to maximum depth while capturing complex patterns.
4. **Prediction:** Average individual tree predictions.

Benefits

- Reduced Overfitting
- Robustness
- Important Feature Identification

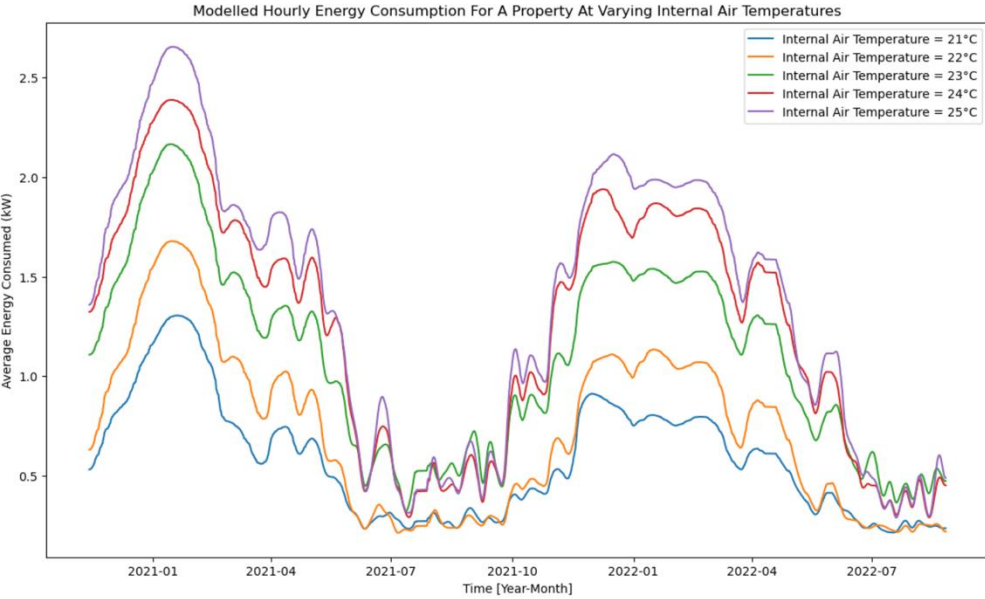
Results & Analysis



On the left, the graphs give the **modelled** and **actual** ASHP energy **consumption series** of a random property chosen from the DESNZ dataset against the **temperature difference** between the indoor and external environments. Above and below, the results are plotted for a model that was trained with the property **included** and **excluded** from the dataset, respectively.

Both plots show good correlation to one another, as well as the temperature difference, which shows that the model is effective in predicting energy consumption from the input features: **Internal Air Temperature, External Air Temperature, SCOP, and Total Energy Consumption.**

Despite good correlation, the **Mean Absolute Percentage Error** for the models are **10.685%** and **32.246%** respectively, suggesting that, as expected, data already present in the training set is modelled with greater accuracy. Future adjustments may be made to **increase the training set** or **random forest size** or **refine the modelling process** for more accurate predictions.



The analysis plot above shows that as predicted, **household air temperature** and **ASHP energy consumption** are proportional, which shows that a consumer should still endeavour to **not excessively heat** the property above a sensible range to **keep costs down**.

Future Work

- Extend model capabilities to account for battery storage and solar power input
- Increase model efficiency and decrease runtime
- Continue to add to dataset to make more accurate predictions

References

- [1] <https://beta.ukdataservice.ac.uk/datacatalogue/studies/study?id=9050#/access-data>
- [2] <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>
- [3] <https://doi.org/10.1016/j.enbuild.2021.111365>
- [4] <https://doi.org/https://doi.org/10.1016/j.applthermaleng.2015.03.026>

GitHub Access

